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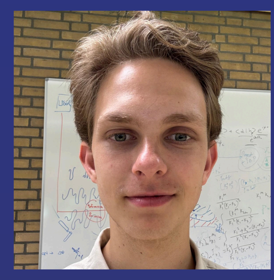
REDUCING NON-LINEARITIES IN HoQI BASED MEASUREMENTS

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Introduction

Gravitational waves are invisible ripples in the fabric of spacetime. Predicted by Albert Einstein in 1916, they travel at the speed of light, stretching and squeezing **spacetime** along their way. They are detected using large laser interferometers, often characterized by their recognizable L-shape. Since they are **transversal** waves, gravitational waves stretch spacetime in one direction perpendicular to the propagation direction, while squeezing it in the other orthogonal direction. As a wave passes through the detector, the lengths of the detector's arms change slightly, causing a **phase shift** between two interfering laser beams. The resulting **interference pattern** implies the passing of a gravitational wave, whose propagation direction can be determined using **triangulation**.

Detecting gravitational waves requires measuring **subatomic fluctuations** in distance, demanding near-perfect mechanical isolation of the optical elements. To achieve this, external **vibrations** are constantly monitored by an **active damping system** of sensors and actuators, operating in tandem with passive isolations to suppress residual motion.

Method

A **prototype** of the aforementioned dampening system is illustrated in Fig. 1 [1] (centered). The setup consists of three arms, all at an angle of nominally 120 degrees to each other. To each of the three arms, two Homodyne Quadrature Interferometers (**HoQIs**) are attached. These HoQIs serve as the high-precision sensors responsible for repressing the system's resonances. They use **laser interferometry** to measure the **position** of a **free-floating test mass** in the plane perpendicular to the associated arm. An optical lay-out of a single HoQI can be seen in Fig. 2 [2]. The three photodetectors (PD1, PD2 and PD3) measure the intensity of the interfering light, converting it to a voltage used to calculate the position of the test mass. Combining the individual measurements of all six HoQIs, we can use the 6x6 **transformation matrix** as given in Eq. (1), to determine the **displacement** of the detector's **center of mass** (CM) in all six degrees of freedom (**X, Y, Z, RX, RY and RZ**).

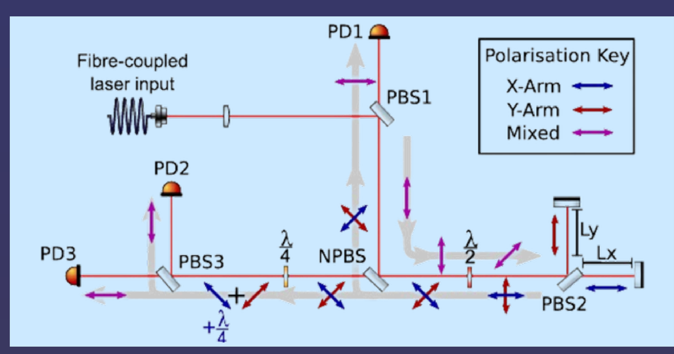

Fig. 2: Optical lay-out of a single HoQI.

$$\begin{pmatrix} -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{2}{3R} & -\frac{1}{3R} & -\frac{1}{3R} \\ 0 & 0 & 0 & 0 & \frac{1}{\sqrt{3}R} & -\frac{1}{\sqrt{3}R} \\ \frac{1}{3R} & \frac{1}{3R} & \frac{1}{3R} & 0 & 0 & 0 \end{pmatrix}$$

Eq. (1): 6x6 transformation matrix used for calculating the position of the CM.

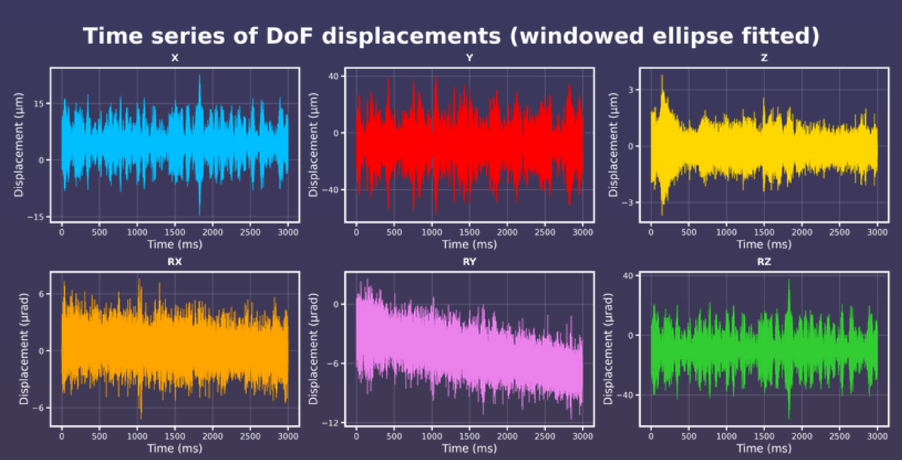

Fig. 3a: Time series of the CM's position in all six degrees of freedom (windowed ellipse fitted).

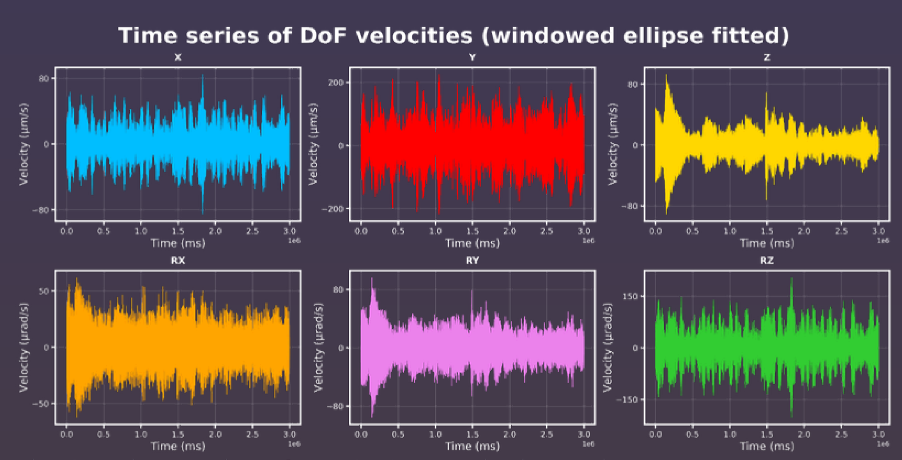

Fig. 3b: Time series of the CM's velocity in all six degrees of freedom (windowed ellipse fitted).

Time Series

Using this transformation matrix, we are able to make a so-called '**time series**' of both the displacement (see Fig. 3a) and the velocity (see Fig. 3b) of the center of mass in all six degrees of freedom. The data used to create these plots come from a 3000-second measurement, with a fixed **sample rate** of exactly 1000 Hz.

These time series can be used to gain insight into the movements of the detector. As will be shown later, this is crucial for the efficient removal of 'non-linearities' from the dataset. **Non-linearities** are systematic distortions where the output signal is no longer directly proportional to the physical input. They are usually caused by either high-amplitude or high-velocity movements, resulting in the infusion of **instrumental noise**.

Improving Linearity

There are multiple ways in which non-linearities could get injected into the measurements. For instance, if the beam splitters in Fig. 2 do not split the laser beams in a perfect **50/50 ratio**, the photodetectors measure deviating intensities, leading to erroneous results. Furthermore, the quarter-wave plate might not delay the oscillation in one direction with exactly **90 degrees**, resulting in a non-perfect sine or cosine. A final cause for non-linearities is a potential **time dependence** of the fringe visibility. **Fringe visibility** is a quantitative measure of the contrast of intereference fringes in any wave-based system. Possible reasons of a time-dependent fringe visibility include imprecise reflection mirrors, a non-perfect 50/50 ratio of the beam splitters, a mechanical flaw in the half-wave plate, and movements in a HoQI's **transverse plane**.

This research project aims to **reduce** the **non-linearities** in **HoQI** based measurements, with a specific focus on the **high-frequency domain**. We have done this using various types of **data analysis**, including ellipse fitting, ASD plotting, and training a machine learning model with the usage of a kNN algorithm.

Fig. 1: Measurement setup at Nikhef, including Cartesian coordinate system.

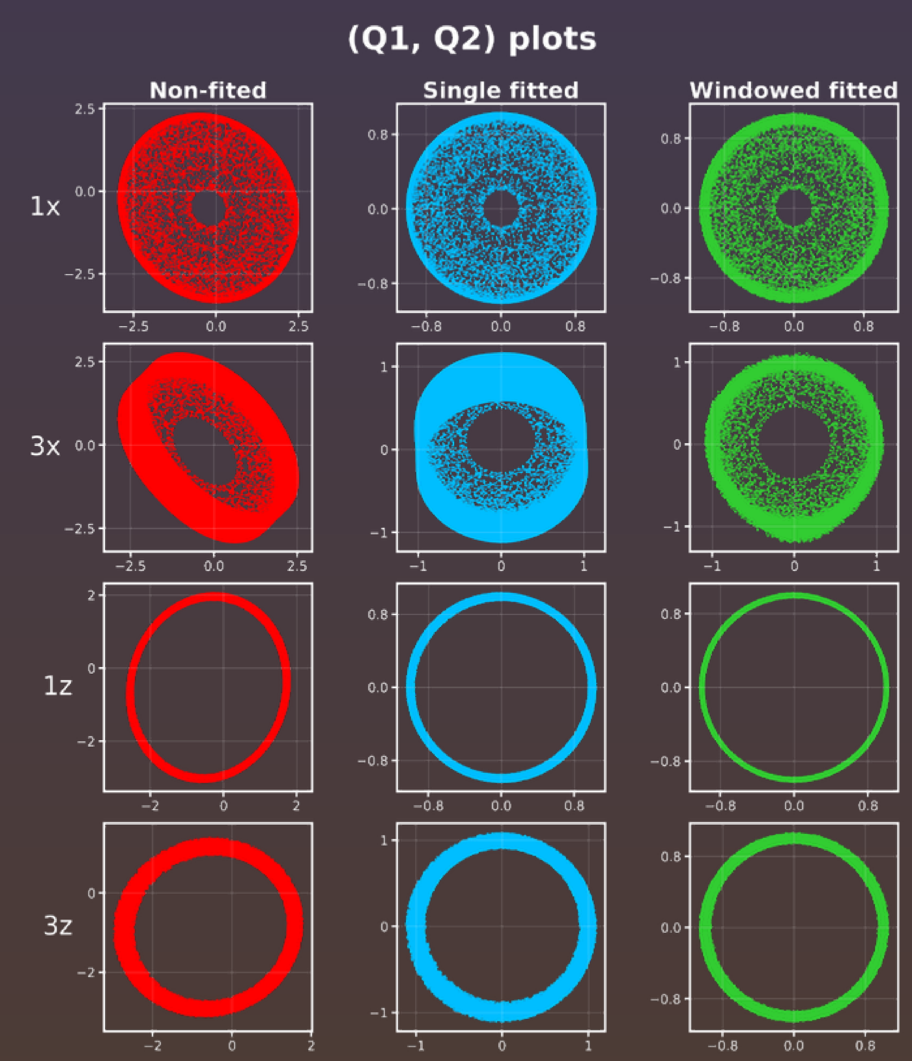

Fig. 4: (Q1, Q2) plots for four HoQIs (1x, 3z, 1z and 3z), comparing (left) the non-fitted, (center) the single ellipse fitted, and (right) the windowed ellipse fitted data.

Ellipse Fitting

The first analytical method we used for reducing the injected non-linearities, is called **'ellipse fitting'**. As can be seen in the **animation** (scan the QR code in the top left corner), plotting the Q1 list of a certain HoQI against the Q2 list of that same HoQI, should - in theory - result in a perfect **circle**. Though, existing **non-linearities** in the measurements cause the data points to fall off this circle, leading to a Lissajous-like figure. In practice, however, the **(Q1, Q2) plots** mainly take on the appearance of an **ellipse** (see Fig. 4a).

The afmorementioned ellipse fitting is a process where we fit one or more ellipses through the plotted data points, after which we use the determined **ellipse parameters** (x_0, y_0, a, b and θ) to map the data points back onto a single circle — the **unit circle**, in our case. An image of the results correseponding to two different ways of ellipse fitting (titled '**single** ellipse fitting' and '**windowed** ellipse fitting') can be seen in Fig. 4b and Fig. 4c respectively.

During the analysis, we also constructed a **time series** of all five ellipse parameters. This revealed that the values of these parameters **fluctuate** over time. Fitting a single ellipse through the (Q1, Q2) data points means every point is **transformed** with the **exact same parameters**, leading to slightly incorrect transformations. Windowed ellipse fitting tries making up for this by fitting an ellipse through the data points for numerous 'windows', using an adjustable window size, step size, and lag. Consequently, each point is **transformed** using **locally optimized parameters**, yielding more accurate results.

ASD Plotting

Amplitude Spectral Density (**ASD**) diagrams are graphical plots that illustrate how the **amplitude** of a signal is distributed across a certain range of **frequencies**. In our case, we applied **Fourier analysis** to the time series of the different HoQI displacements, decomposing the signals into their constituent sine waves. Fig. 6 shows the result for HoQI 3x for all **three cases**: the non-fitted, the single ellipse fitted, and the windowed ellipse fitted data.
ASD diagrams are used to determine the **peak frequencies** of oscillation systems, allowing scientists to optimize their measurement setup by mitigating **resonances** and **bending modes**.

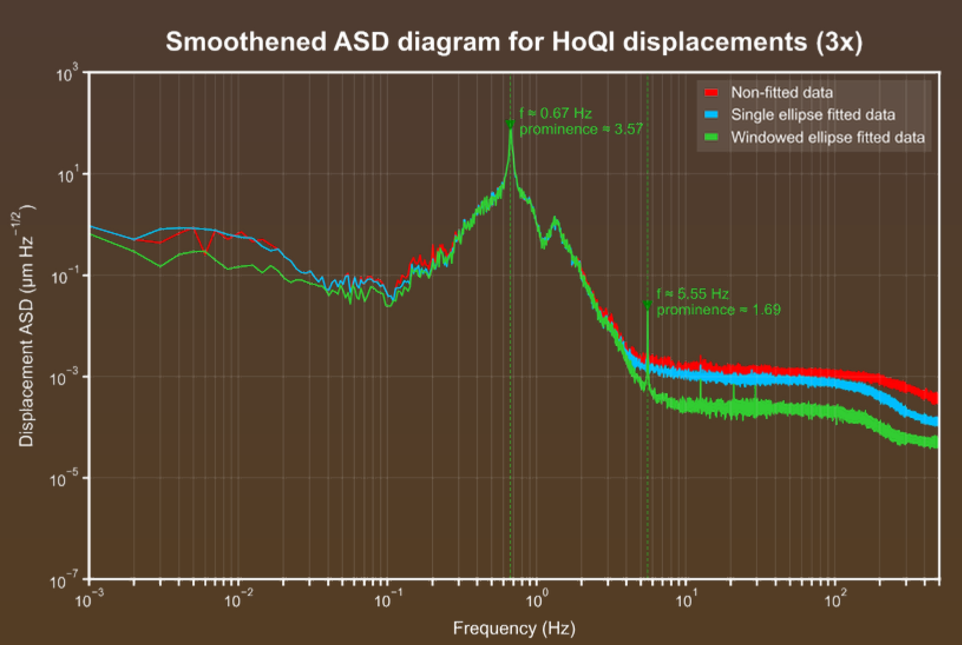

Fig. 6: Smoothened ASD diagram for the displacments of HoQI 3x, comparing the three different data processing approaches.

kNN Algorithm

One of our hypotheses was that the ellipse parameters depend on the position of the test mass in the transverse plane of the relevant HoQI. To test this hypothesis, a **kNN algorithm** was used for training a **machine learning model** to check whether it could accurately **predict** future parameters (see Fig. 7). We compared a **static** kNN model, whose training data consists of the first 80,000 data points, to an **active** kNN model, whose training data consists of constantly updating data points.

Both kNN models mostly achieved **RMSE values** below the baseline (see Tab. 1), indicating that the ellipse parameters can indeed be predicted from the position in the transverse plane. The active model performed **slightly better** than the static model for most of the data segments we tested them on. This suggests that the **underlying patterns** remain relatively stable over time, although temporal **non-linear effects** may still be present.

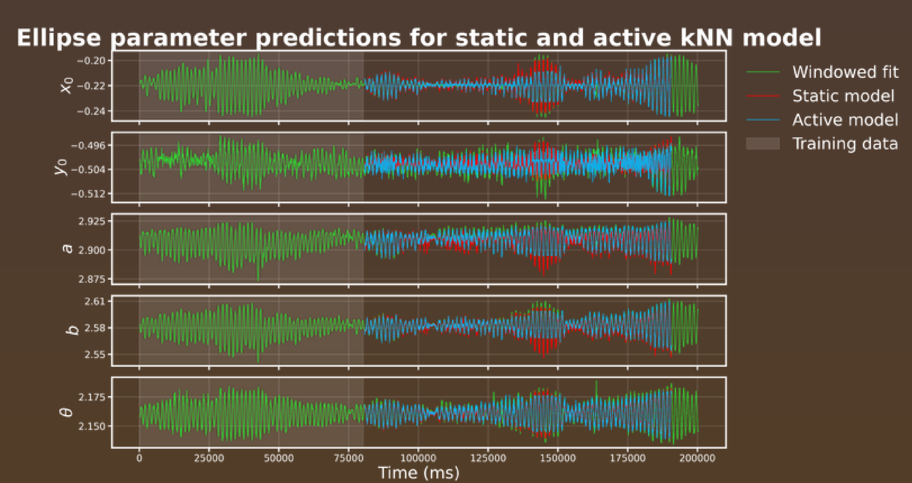

Fig. 7: Ellipse parameter predictions using a kNN algorithm, showed for both an active and a static model.

	RMSE (a.u.)	Gain (%)
Baseline	4.762 ± 2.725	–
Static	3.301 ± 1.219	30.7
Active	2.991 ± 1.518	38.9

Tab. 1: kNN results. Gain relative to baseline.

Conclusion

As shown in the **(Q1, Q2) plots** in Fig. 3, applying ellipse fitting methods yields **significant improvements** compared to the raw, non-fitted data. Although windowed ellipse fitting results in additional progress relative to single ellipse fitting, the increase in quality is **less pronounced**.

The **ASD diagram**, as displayed in Fig. 6, further **reinforces** this statement. Especially in the high-frequency domain, a **substantial reduction** in **noise levels** has been achieved. This has led to more clearly visible peak frequencies, thereby enabling the precise detection and subsequent removal of specific bending modes.
Finally, the **kNN results** shown in Fig. 7 **confirm** our hypothesis regarding the dependence of the ellipse parameters on the test mass' position in the transverse plane. This dependence is subject to minor **temporal fluctuations**, driven by underlying, independently varying **non-linearities**.

References

[1] A. L. Mitchell. "HoQI Low-noise Interferometric Sensors for the Future of Gravitational Wave Detectors". PhD thesis. Department of Physics and Astronomy, Vrije Universiteit Amsterdam (2024).
[2] S. J. Cooper e.a. "A compact, large-range interferometer for precision measurement and inertial sensing". In: Classical and Quantum Gravity 35 (2018), p. 095007-1-095007-9.